nanoGPT hyperparameter experiments

Group 7 - Member 2 – Jonathan Fugitt | block_size = 64 | n_layer = 6

1. Introduction

For my part of Assignment 11, I tested 32 different nanoGPT configurations using the Tiny Shakespeare dataset. I kept the context length at 64 characters and the number of transformer layers at 6, which were the settings assigned to Member 2. I changed the remaining five hyperparameters to see how they affected training and validation loss.

2. Experimental setup

Setting	Values
Fixed settings	block_size = 64; n_layer = 6
Attention heads (n_head)	4, 8
Embedding size (n_embd)	128, 256
Batch size	8, 16
Training iterations (max_iters)	1,000, 2,000
Dropout	0.1, 0.2

I ran the experiments in Google Colab using the nanoGPT code and the same general notebook setup as Member 3. Each run started as a new model and saved a training log, loss history, and generated-text sample. The notebook also records the runtime and approximate model size.

3. Main results

Result	Run	Validation loss	Train loss	Time
Lowest validation loss	g2_31	1.7522	1.5913	38.10 sec
Highest validation loss	g2_02	2.2718	2.2809	23.78 sec

The best run used 8 attention heads, 256 embedding dimensions, batch size 16, 2,000 iterations, and dropout 0.1. The worst run used 4 heads, 128 dimensions, batch size 8, 1,000 iterations, and dropout 0.2. The difference in validation loss between those runs was 0.5196.

4. Average validation loss by setting

Parameter	Lower	Avg. loss	Higher	Avg. loss
Heads	4	2.0373	8	2.0251
Embedding dimensions	128	2.1147	256	1.9476
Batch size	8	2.0928	16	1.9695
Iterations	1000	2.1049	2000	1.9574
Dropout	0.1	1.9964	0.2	2.0660

The biggest average improvement came from increasing the embedding size. More iterations and a larger batch size also helped. Eight attention heads were only slightly better than four. Dropout 0.1 did better than 0.2 in these runs.

5. Results for all 32 configurations

All models used block_size = 64 and n_layer = 6. H = attention heads; E = embedding dimensions; B = batch size. Lower validation loss is better.

Run	H	E	B	Iters	Drop	Train	Val.	Sec
g2_01	4	128	8	1000	0.1	2.2168	2.2211	28.0
g2_02	4	128	8	1000	0.2	2.2809	2.2718	23.8
g2_03	4	128	8	2000	0.1	2.0290	2.0826	33.8
g2_04	4	128	8	2000	0.2	2.1238	2.1512	31.2
g2_05	4	128	16	1000	0.1	2.0483	2.0947	18.0
g2_06	4	128	16	1000	0.2	2.1263	2.1513	18.2
g2_07	4	128	16	2000	0.1	1.8542	1.9787	33.0
g2_08	4	128	16	2000	0.2	1.9592	2.0510	32.4
g2_09	4	256	8	1000	0.1	1.9975	2.0635	18.3
g2_10	4	256	8	1000	0.2	2.0933	2.1353	18.4
g2_11	4	256	8	2000	0.1	1.7558	1.8956	32.6
g2_12	4	256	8	2000	0.2	1.8573	1.9698	32.2
g2_13	4	256	16	1000	0.1	1.8078	1.9431	20.6
g2_14	4	256	16	1000	0.2	1.8953	2.0061	20.8
g2_15	4	256	16	2000	0.1	1.5950	1.7532	35.8
g2_16	4	256	16	2000	0.2	1.6837	1.8271	36.1
g2_17	8	128	8	1000	0.1	2.1917	2.1845	18.8
g2_18	8	128	8	1000	0.2	2.2665	2.2555	18.4
g2_19	8	128	8	2000	0.1	1.9976	2.0532	30.7
g2_20	8	128	8	2000	0.2	2.1104	2.1390	30.9
g2_21	8	128	16	1000	0.1	2.0193	2.0628	18.9
g2_22	8	128	16	1000	0.2	2.1147	2.1418	18.9
g2_23	8	128	16	2000	0.1	1.8282	1.9600	31.7
g2_24	8	128	16	2000	0.2	1.9450	2.0362	32.0
g2_25	8	256	8	1000	0.1	1.9928	2.0684	18.0
g2_26	8	256	8	1000	0.2	2.0941	2.1426	19.4
g2_27	8	256	8	2000	0.1	1.7519	1.8904	32.5
g2_28	8	256	8	2000	0.2	1.8561	1.9606	32.1
g2_29	8	256	16	1000	0.1	1.8151	1.9379	20.5
g2_30	8	256	16	1000	0.2	1.8882	1.9982	20.3
g2_31	8	256	16	2000	0.1	1.5913	1.7522	38.1
g2_32	8	256	16	2000	0.2	1.6704	1.8183	36.6

Source: results/member2/results.csv, included in Group7_Member2_Results.zip. All 32 run IDs appear once. Total recorded run time: 851.07 seconds (about 14.2 minutes).

6. Loss curves and generated text

This graph compares the first run, the run with the highest final validation loss, and the run with the lowest final validation loss. The 2,000-iteration run also had more training steps, so this is not a comparison of only one hyperparameter.

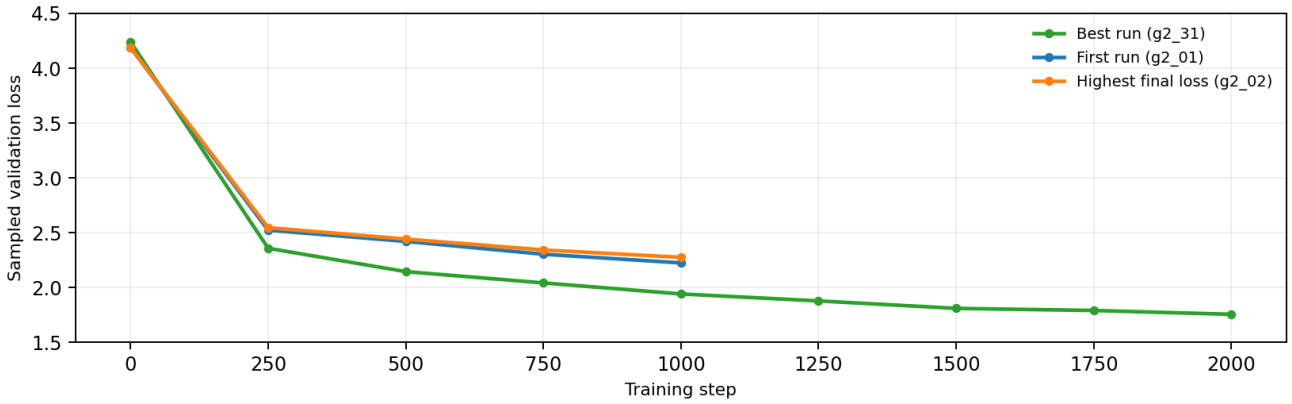


Figure 1. Sampled validation loss during training. Data comes from each run's saved loss_history.csv.

Generated text

Both models generated text from the same "ROMEO:" prompt, with temperature 0.8, top_k = 40, and seed 1337. These are short excerpts from the actual saved samples.

g2_31 | best final loss	g2_02 | highest final loss
ROMEO: You before will and is the servaited bube to take Our some? Takes: Whith foul him to behed lady, ever you had for a will our	ROMEO: Al the Ror we wild to on, be mad and bube ale.

7. Discussion and conclusion

The models with lower validation loss generally produced more recognizable words and dialogue-like formatting, but even the best sample still made up words and did not make much sense as a conversation. The larger embedding size and longer training runs gave the clearest improvements in this experiment. Increasing batch size also helped, although that changes how many characters the model processes at each step.

Each configuration was trained once, so these results show what happened in this set of runs, not necessarily which settings are always best. My validation numbers come from sampled batches; Member 1 used a full-validation calculation, so I would not compare those numbers directly. The notebook and ZIP contain the full logs, graphs, and generated text for all 32 runs.